COLLEGE HELPDESK CHATBOT

Mini Project Report

Technology: Python (if-elif-else conditions)

Project type: Rule-based console chatbot

1. Introduction

The College Helpdesk Chatbot is a simple rule-based program that responds to common student questions. It checks the words entered by the user and chooses a prepared response using Python conditional statements. It uses only Python's built-in features.

2. Aim

To develop a rule-based chatbot in Python using if-elif-else conditions that answers basic questions related to college services.

3. Objectives

• Accept text input from the user.

• Identify common keywords in the input.

• Respond to questions about admission, courses, fees, exams, library, timings, and contact details.

• Continue the conversation until the user types 'bye'.

4. Tools and Requirements

• Python 3
• Any code editor or Python IDLE
• Terminal/command prompt
• No external libraries, internet connection, API key, or database is required to run locally.

5. Working

The program displays a welcome message and repeatedly asks the user to enter a message. It converts the input to lowercase, then checks it against predefined rules. When a rule matches, the chatbot prints the corresponding response. If no rule matches, it displays a fallback response. The loop ends when the user types 'bye', 'exit', or 'quit'.

6. Sample Chatbot Conversations

```
Bot: Hello! Welcome to the college helpdesk. How can I help you?\nYou: hi\nBot: Hello! Welcome to the

You: How can I get admission?\nBot: For admission, keep your previous marksheet, ID proof, photograph

You: Tell me about courses\nBot: Colleges may offer diploma, undergraduate, and postgraduate programs

You: What are the fees?\nBot: Fee amounts depend on the course and semester. Please check the latest

You: exam timetable\nBot: Please check the official college notice board or student portal for the la

You: library books\nBot: The library can help you find textbooks and reference books. Ask the librari

You: thank you\nBot: You're welcome! Happy to help.\nYou: bye\nBot: Goodbye! Best of luck with your s
```

7. Advantages

• Simple and easy to understand for beginners.
• Uses only Python and no third-party libraries.

• Gives quick responses to supported questions.
• Can be extended by adding more if-elif rules.

8. Limitations

This chatbot does not understand natural language like an AI model. It matches keywords, so differently phrased questions may not be recognized. Its answers are sample guidance; students should confirm official fees, schedules, and requirements with their college.

9. How to Run

Save the code as **college_chatbot.py**. Open a terminal in the folder and run **python college_chatbot.py**. Enter a question when the 'You:' prompt appears. Type **bye** to stop the program.

10. Conclusion

The College Helpdesk Chatbot demonstrates the basic concept of a rule-based chatbot using Python conditional statements and a loop. It provides a simple way to answer frequently asked college-related questions and can be improved with additional rules in the future.

Submission note: This is a console chatbot. The Python file runs locally without setup beyond Python 3. A GitHub repository can host the code, but it does not itself provide a browser-based interactive live chatbot.